

STATISTICAL BUSINESS ANALYSIS REPORT

Name: Jaya Krishna Pediredla

Project: Statistical Business Analysis

Internship: The Developers Arena

Project Objective:

To analyze business sales data using descriptive statistics, hypothesis testing, confidence intervals, correlation analysis and regression analysis.

Dataset Description:

Columns:

Date

Product

Quantity

Price

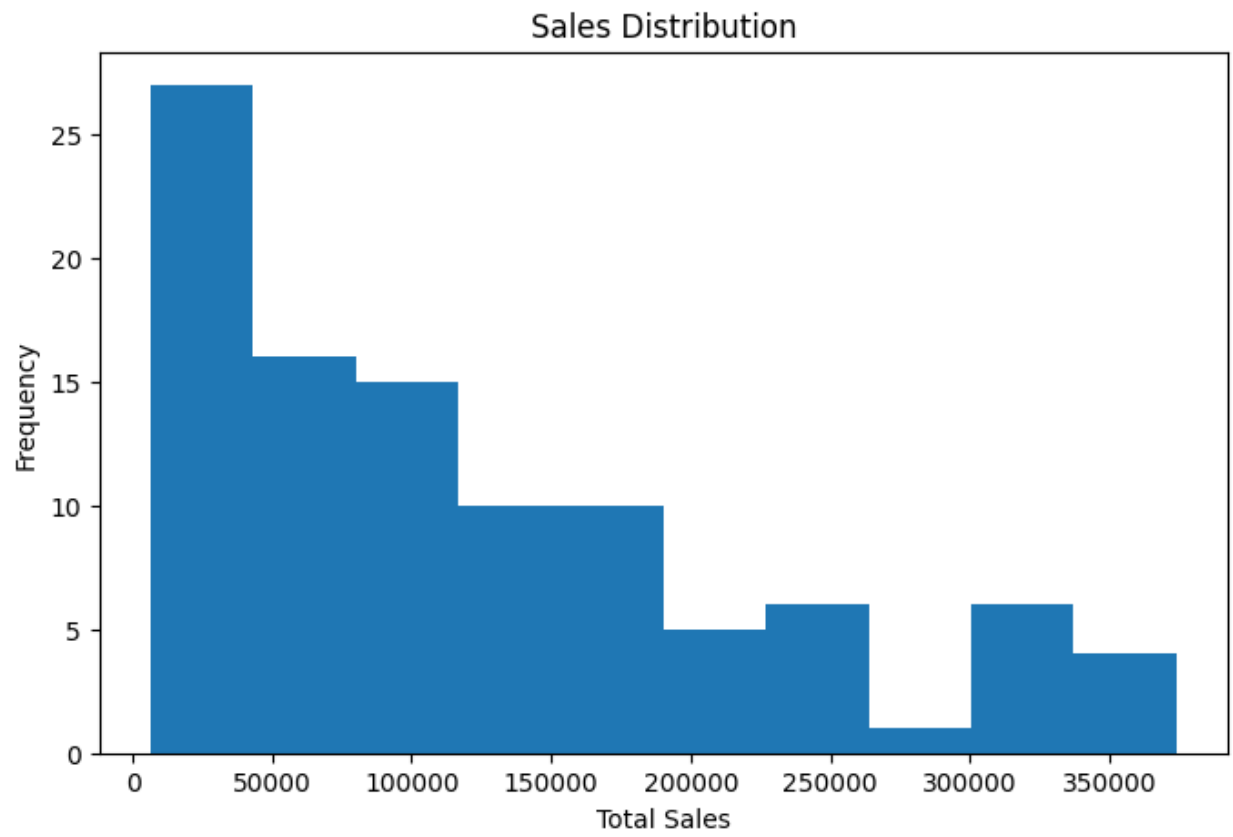
Customer_ID

Region

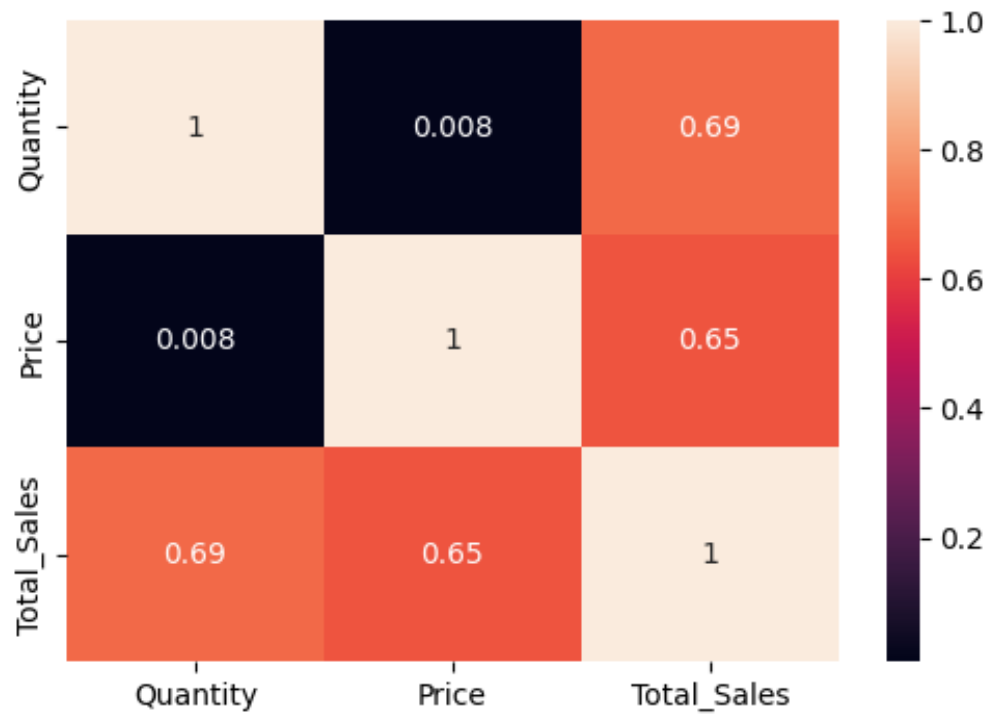
Total_Sales

sales.describe()			
[6]	✓	0.0s	Python
...			
	Quantity	Price	Total_Sales
count	100.000000	100.000000	100.000000
mean	4.780000	25808.510000	123650.480000
std	2.588163	13917.630242	100161.085275
min	1.000000	1308.000000	6540.000000
25%	2.750000	14965.250000	39517.500000
50%	5.000000	24192.000000	97955.500000
75%	7.000000	38682.250000	175792.500000
max	9.000000	49930.000000	373932.000000

descriptive statistics



histogram



heatmap

```
east = sales[sales["Region"]=="East"]["Total_Sales"]
west = sales[sales["Region"]=="West"]["Total_Sales"]

t_stat, p_value = stats.ttest_ind(east, west)

print("T Statistic:", t_stat)
print("P Value:", p_value)
```

[16] ✓ 0.0s

```
... T Statistic: 2.0202108796829767
    P Value: 0.049619032636626016
```

hypothesis testing outputs



```
north = sales[sales["Region"]=="North"]["Total_Sales"]
south = sales[sales["Region"]=="South"]["Total_Sales"]

t_stat, p_value = stats.ttest_ind(north, south)

print("T Statistic:", t_stat)
print("P Value:", p_value)
```

Rectangular Snip

[15]

✓ 0.0s

Python

...

```
T Statistic: 0.1303316283399723
P Value: 0.8967974805172009
```

```
east = sales[sales["Region"]=="East"]["Total_Sales"]
north = sales[sales["Region"]=="North"]["Total_Sales"]

t_stat, p_value = stats.ttest_ind(east, north)

print("T Statistic:", t_stat)
print("P Value:", p_value)
```

[14]

✓ 0.0s

Python

...

```
T Statistic: -0.35710677600784724
P Value: 0.7226811638407538
```

```
mean = sales["Total_Sales"].mean()

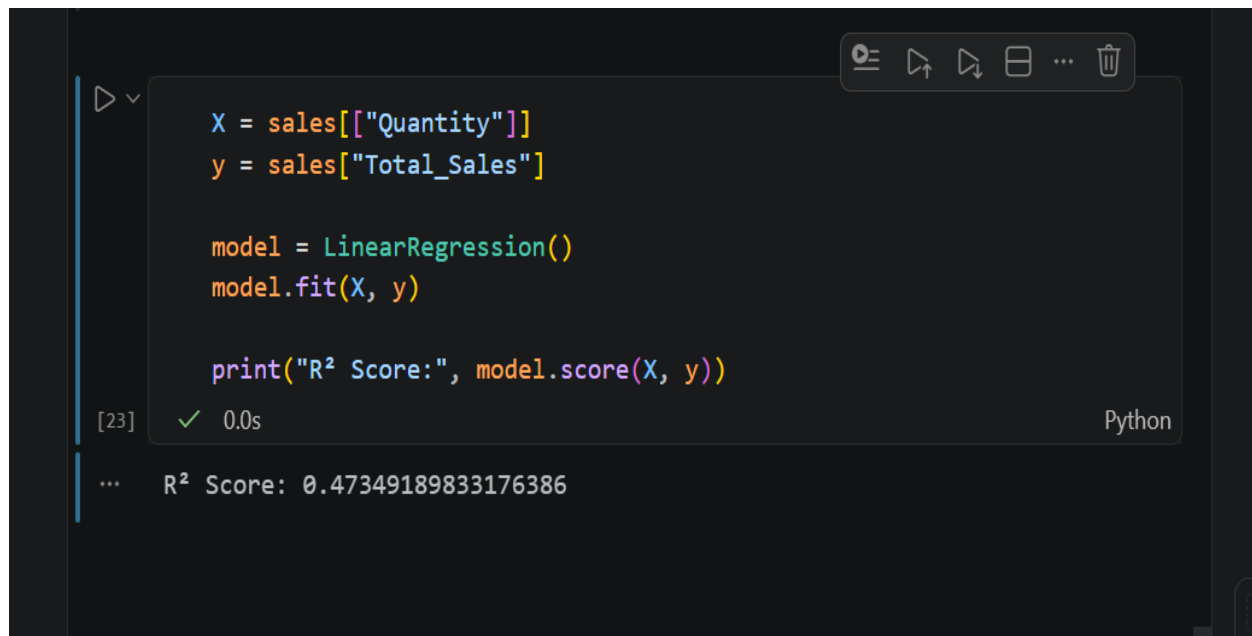
confidence_interval = stats.t.interval(
    0.95,
    len(sales)-1,
    loc=mean,
    scale=stats.sem(sales["Total_Sales"])
)

print(confidence_interval)
```

[22] ✓ 0.0s Python

... (np.float64(103776.3476708068), np.float64(143524.6123291932))

confidence interval output



A screenshot of a Jupyter Notebook interface. The code cell contains the following Python code: `X = sales[["Quantity"]]`, `y = sales["Total_Sales"]`, `model = LinearRegression()`, `model.fit(X, y)`, and `print("R2 Score:", model.score(X, y))`. The output cell shows the execution result: `R2 Score: 0.47349189833176386`. The interface includes a toolbar with icons for running, stepping through, and saving code, as well as a language selector set to 'Python'.

```
X = sales[["Quantity"]]
y = sales["Total_Sales"]

model = LinearRegression()
model.fit(X, y)

print("R2 Score:", model.score(X, y))
```

[23] ✓ 0.0s Python

... R² Score: 0.47349189833176386

regression output

R² Score = 0.4735

Business Insights:

1. Quantity has a positive impact on sales.
2. Regional sales performance can be compared using hypothesis testing.
3. Correlation analysis helps identify relationships between variables.
4. Statistical methods support data-driven decision making.

Conclusion:

The project successfully demonstrated the application of statistical techniques for business data analysis. The findings provide meaningful insights that can support business decision-making.